



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Algebra Toolkit

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Your notes

Algebraic Notation

Algebraic Notation

What is algebra?

- Algebra is a topic in mathematics that uses **letters** to represent **general** (or **unknown**) **numbers**
 - x and y are two unknown numbers
 - More information is needed to find their values
- Letters are also called **variables**

How do I write calculations in algebra?

- Writing mathematical ideas in letters is called using **algebraic notation**
- You can use $+$ and $-$ to mean **add** and **subtract**
 - $a + b$
 $c + d - e$
- However **no** symbol is used for **multiplication**
 - ab (means $a \times b$)
 $3ab$ (means $3 \times a \times b$)
- Fractions** are used for **division**
 - $\frac{a}{b}$ (means $a \div b$)
- You can combine the ideas above
 - $ab + \frac{c}{3}$ (means $a \times b + c \div 3$)
 - The **order of operations** for numbers still works
 - work out $a \times b$ and $c \div d$ before adding them together
- Powers (indices)** and **roots** are the same as with numbers
 - a^2 means $a \times a$
 $4a^2$ means $4 \times a^2$

- With the order of operations, a^2 happens before multiplying by 4
- $\sqrt{a}$ means the square root of a
- **Brackets** work in the same way as they do with numbers
 - $3(a + b)$ means $3 \times (a + b)$
 - Add the a and b first, then multiply the result by 3



Worked Example

Raheem is playing a game and starts with q points.
He then earns a further 6 points.

(a) Use algebra to write down the total number of points that Raheem has.

Raheem has $q + 6$ points

The game then decides to double his total number of points.

(b) Use algebra to write down the number of points that Raheem now has.

Raheem has $2(q + 6)$ points



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Algebraic Vocabulary

Algebraic Vocabulary

What is a term?

- A **term** is either:
 - a letter (**variable**) on its own, or a variable raised to a power
 - For example, x or x^2
 - a number on its own
 - For example, 20
 - These are also called **constants** as they can't change value
 - or a number **multiplied** by a letter
 - For example, $5x$
- The number in front of a letter is called a **coefficient**
 - The coefficient of x in $6x$ is 6
 - The coefficient of y in $-5y$ is -5
- Terms can include **powers** and more than one letter,
 - $6xy$
 - $4x^2$
 - ab^3c

What is a factor?

- A **factor** is any number or letter that **divides** a term **exactly**
 - There is no remainder
 - The factors of $3x$ are 1, 3, x and $3x$
 - The factors of $4xy$ are 1, 2, 4, x , $2x$, $4x$, y , $2y$, $4y$, xy , $2xy$ and $4xy$
- A term can be separated into **factors** that **multiply** together to give that term
 - Two factors of $5x$ are 5 and x



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- $5 \times x = 5x$
- To **factorise** means to write something as a **multiplication of factors**
- When comparing two terms, a **common factor** is one that divides both
 - Compare $6xy$ with $4x$
 - Common factors are 1, 2, x and $2x$
 - The **highest** (or greatest) common factor is $2x$

What is an expression?

- An **expression** is an algebraic statement that does **not** have an **equals sign**
 - There is nothing to solve
- An expression is made by adding, subtracting, multiplying or dividing **terms**
 - $2x + 5y$
 - $b^2 - 2cd$
 - $\frac{6y}{5t}$
 - A single term is still an expression
- Expressions can be **simplified** (made easier)
 - $x + x + x$ simplifies to $3x$

What is an equation?

- An **equation** is an algebraic statement with an **equals sign** between a **left-hand side** and a **right-hand side**
 - Both sides are **equal** in value
 - For example, if $2x$ has the same value as 10, then $2x = 10$
- An equation can be **solved** by finding the values of the letters that make both sides equal
 - The equation $2x = 10$ is solved when x has the value of 5
 - $x = 5$ is called the **solution**

What is a formula?

- A **formula** is a rule, definition or relationship between different quantities, written in shorthand using **letters**



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- For example, weight, w , is mass, m , multiplied by gravitational acceleration, g
 - The formula is $w = mg$
- It is common to **substitute** numbers into a formula
 - But a formula on its own cannot be solved
- To turn a **formula** into an **equation**, more information is needed
 - For example, if $w = 50$ and $m = 5$, the formula $w = mg$ becomes the equation $50 = 5g$



Examiner Tips and Tricks

- Knowing the differences between an **expression**, an **equation** and a **formula** will help you to understand the wording of exam questions.



Worked Example

(a) From the list below, write down

- (i) an expression,
(ii) an equation.

$$2x + 5 = 4 \quad 7x - 9 \quad x = vt - w$$

(i) An expression does not have an equals sign

$7x - 9$ is the expression

(ii) An equation has an equals sign and can be solved

$2x + 5 = 4$ is the equation

(b) If $x = 10$, $v = 2$ and $w = 3$, use the formula shown to write an equation in t .

$x = vt - w$ is the formula shown (a group of different quantities forming a relationship)

Substitute $x = 10$, $v = 2$ and $w = 3$ into the formula

$$10 = 2t - 3$$



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Substitution

Substitution

What is substitution?

- **Substitution** means **replacing** a letter (variable) in a formula with a given **number**

How do I substitute numbers into a formula?

- Write down the **formula**
- **Replace** (substitute) the letters in the formula with the given numbers
 - If substituting in a **negative number**, it is important to put **brackets** around it
 - For example, (-3)
- **Simplify** any numerical calculations
- **Calculate** the final value
- Sometimes the result is an **equation** which you can then **solve**



Examiner Tips and Tricks

- On your calculator, don't forget to type out **brackets** around any substituted **negative** numbers!



Worked Example

(a) Find the value of the expression $2x(x + 3y)$ when $x = 2$ and $y = -4$.

Substitute the numbers given

Use brackets $()$ around negative numbers

$$2 \times 2 \times (2 + 3 \times (-4))$$

Complete the calculation

Show every step of working, following the order of operations correctly



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$$\begin{aligned} &= 2 \times 2 \times (2 - 12) \\ &= 2 \times 2 \times (-10) \\ &= 4 \times (-10) \end{aligned}$$

-40

(b) The formula $P = 2l + 2w$ is used to find the perimeter, P , of a rectangle of length l and width w .

Given that the rectangle has a perimeter of 20 cm and a width of 4 cm, find its length.

Substitute the values you are given for P and w into the formula

$$20 = 2 \times l + 2 \times 4$$

Simplify

$$20 = 2l + 8$$

Solve the resulting equation to find the value of l

Start by subtracting 8 from both sides

$$12 = 2l$$

Divide both sides by 2

$$l = 6 \text{ cm}$$



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Collecting Like Terms

Collecting Like Terms

What happens if there is more than one term?

- **Terms** can be **added** and **subtracted**
 - The **numbers** in front of the letters are called **coefficients**
- Each term has a positive or negative **sign** in front
 - In $2x - 3y$ the sign of the x term is positive and the sign of the y term is negative
- **Subtractions** can be thought of as **adding a negative**
 - $2x - 3y$ is the same as $2x + (-3y)$
 - Just like $20 - 30$ is the same as $20 + (-30)$
- The **order** of two terms can be **swapped**, but the **signs** must **move** with their terms
 - $2x - 3y$ is the same as $-3y + 2x$
 - A **plus** is now needed in front of the $2x$
 - Just like $20 - 30$ is the same as $-30 + 20$
- If **no number** appears in front of a term, then its number is **1**
 - x is the same as $1x$

What is a like term?

- **Like terms** are terms with exactly the same **letters** and **powers**
 - The **numbers** in front can be **different**
 - For example:
 - $2x$ and $3x$
 - $4x^2$ and $6x^2$
 - $5xy$ and $-7xy$
 - These are **not** like terms:
 - $2x$ and $3y$ (different letters)

- $4x^2$ and $6x^4$ (different powers)
- $5xy$ and $7xyz$ (different letters)
- Remember **multiplication** can be done in **any order**
 - xy and yx are **like** terms
 - So are $2xy$ and $3yx$

How do I collect like terms?

- **Collecting like terms** means simplifying by **adding** or **subtracting** the **numbers** in front
 - $2x + 3x$ becomes $5x$
 - $4y - 10y$ becomes $-6y$
 - A negative sign is needed here
- If there are **different types** of like terms, **collect** them **separately**
 - For $2x + 4y + 5x - 3y$
 - Collecting the x 's gives $2x + 5x = 7x$
 - Collecting the y 's gives $4y - 3y = y$
 - The answer is $7x + y$



Examiner Tips and Tricks

- Don't leave terms like $1x$ in your final answer in an exam - always simplify them to just x .



Worked Example

Simplify

$$8a - 5b - 6a + 4b$$

Collect the a terms first

$$8a - 6a = 2a$$

Then collect the b terms

Don't forget the minus sign in front of the $5b$



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Add together the two answers

$$-5b + 4b = -b$$

$$2a + -b$$

Simplify the signs

$$2a - b$$